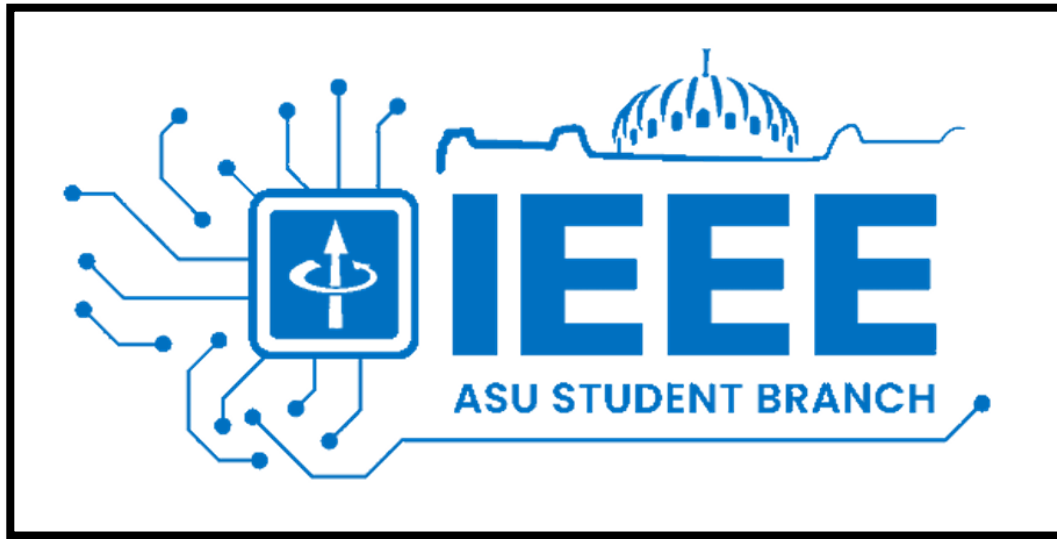




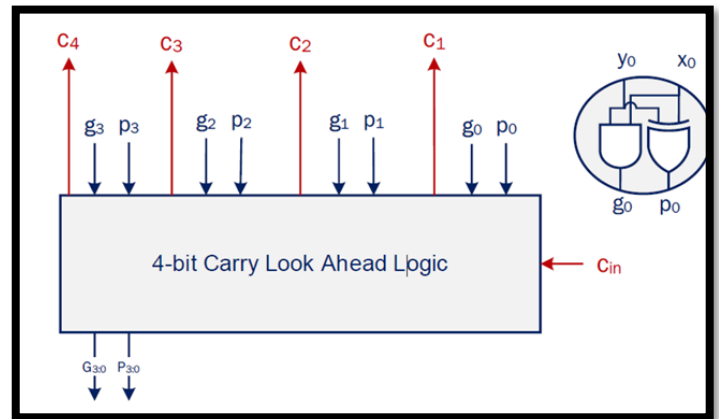
=====IEEE – Computer Arithmetic Workshop=====

=====Assignment(1)=====

> 64-bit Carry Look ahead Adder <

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Step 1: 4bit CLA



-RTL:

```
1 module Carry_LookAhead_4bit(  
2     input Cin, g0, p0, g1, p1, g2, p2, g3, p3,  
3     output C1, C2, C3, Cout, G3_0, P3_0  
4 );  
5  
6     assign C1 = g0 | (p0 & Cin);  
7     assign C2 = g1 | (p1 & g0) | (p1 & p0 & Cin);  
8     assign C3 = g2 | (p2 & g1) | (p2 & p1 & g0) | (p2 & p1 & p0 & Cin);  
9     //OPTION 1: More Area More Speed  
10    //assign Cout = g3 | (p3 & g2) | (p3 & p2 & g1) | (p3 & p2 & p1 & g0) | (p3 & p2 & p1 & p0 & Cin);  
11    //OPTION 2: Less Area Less Speed (As in slides)  
12    assign Cout = g3 | (C3 & p3);  
13  
14  
15  
16    assign G3_0 = g3 | (p3 & g2) | (p3 & p2 & g1) | (p3 & p2 & p1 & g0);  
17    assign P3_0 = p3 & p2 & p1 & p0;  
18  
19 endmodule
```

Step 2.1: 16 bit CLA -RTL:

```
1 module Carry_LookAhead_16bit( //using 4 bit CLA(s)
2     input [15:0] g,p, //ready from top module later
3     input Cin,
4     output [15:0] C,
5     output Cout,
6     output G15_0, P15_0
7 );
8
9
10 //Carry Network=====
11 //wire [15:0] C; //wont use C[0] , C[4] , C[8] ..... in first level
12 wire G3_0 ,P3_0 ,G7_4 ,P7_4 ,G11_8 ,P11_8 ,G15_12 ,P15_12;
13 //assign C[0] = Cin;
14
15 //Layer1=====
16 //0-3
17 Carry_LookAhead_4bit CLA_4bit_0_3
18 (
19     .Cin(Cin),
20     .g0(g[0]), .g1(g[1]), .g2(g[2]), .g3(g[3]),
21     .p0(p[0]), .p1(p[1]), .p2(p[2]), .p3(p[3]),
22     .C1(C[1]), .C2(C[2]), .C3(C[3]), .Cout(), //no need to take C4 / Cout we will predict it
23     .G3_0(G3_0), .P3_0(P3_0)
24 );
25
26 //4-7
27 Carry_LookAhead_4bit CLA_4bit_4_7
28 (
29     .Cin(C[4]),
30     .g0(g[4]), .g1(g[5]), .g2(g[6]), .g3(g[7]),
31     .p0(p[4]), .p1(p[5]), .p2(p[6]), .p3(p[7]),
32     .C1(C[5]), .C2(C[6]), .C3(C[7]), .Cout(), //no need to take C8 / Cout we will predict it
33     .G3_0(G7_4), .P3_0(P7_4)
34 );
35
36 //8-11
37 Carry_LookAhead_4bit CLA_4bit_8_11
38 (
39     .Cin(C[8]),
40     .g0(g[8]), .g1(g[9]), .g2(g[10]), .g3(g[11]),
41     .p0(p[8]), .p1(p[9]), .p2(p[10]), .p3(p[11]),
42     .C1(C[9]), .C2(C[10]), .C3(C[11]), .Cout(), //no need to take C12 / Cout we will predict it
43     .G3_0(G11_8), .P3_0(P11_8)
44 );
45
46 //12-15
47 Carry_LookAhead_4bit CLA_4bit_12_15
48 (
49     .Cin(C[12]),
50     .g0(g[12]), .g1(g[13]), .g2(g[14]), .g3(g[15]),
51     .p0(p[12]), .p1(p[13]), .p2(p[14]), .p3(p[15]),
52     .C1(C[13]), .C2(C[14]), .C3(C[15]), .Cout(), //no need to take C16 / Cout we will get it from next
53     .G3_0(G15_12), .P3_0(P15_12)
54 );
55
56 //Layer2=====
57 Carry_LookAhead_4bit CLA_4bit_for_Carry_Prediction
58 (
59     .Cin(Cin),
60     .g0(G3_0), .g1(G7_4), .g2(G11_8), .g3(G15_12),
61     .p0(P3_0), .p1(P7_4), .p2(P11_8), .p3(P15_12),
62     .C1(C[4]), .C2(C[8]), .C3(C[12]), .Cout(Cout), //Get C16 / Cout from this level
63     .G3_0(G15_0), .P3_0(P15_0)
64 );
65
66
67 endmodule
```

Step 2.2: 64 bit CLA -RTL:

```
1 module Carry_LookAhead_64bit( //using 16 bit CLA(s) || This is top Module
2     input [63:0] A,B,
3     input Cin,
4     output [63:0] S,
5     output Cout
6 );
7
8 //1.G, P Generation Logic=====
9 wire [63:0] g, p;
10 assign g = A & B;
11 assign p = A ^ B;
12 //=====
13
14
15
16 //2.Carry Network=====
17 wire [63:0] C; //wont use C[0] , C[16] , C[32] ..... in first level
18 wire G15_0 ,P15_0 ,G31_16 ,P31_16 ,G47_32 ,P47_32 ,G63_48 ,P63_48;
19 assign C[0] = Cin;
20
21 //Layer1=====
22 //0-15
23 Carry_LookAhead_16bit CLA_16bit_0_15
24 (
25     .Cin(Cin),
26     .g(g[15:0]), .p(p[15:0]),
27     .C(C[15:0]),.Cout(), //won't use C16 / Cout we will predict it
28     .G15_0(G15_0), .P15_0(P15_0)
29 );
30
31 //16-31
32 Carry_LookAhead_16bit CLA_16bit_16_31
33 (
34     .Cin(C[16]),
35     .g(g[31:16]), .p(p[31:16]),
36     .C(C[31:16]),.Cout(), //won't use C32 / Cout we will predict it
37     .G15_0(G31_16), .P15_0(P31_16)
38 );
39
40 //32-47
41 Carry_LookAhead_16bit CLA_16bit_32_47
42 (
43     .Cin(C[32]),
44     .g(g[47:32]), .p(p[47:32]),
45     .C(C[47:32]),.Cout(), //won't use C48 / Cout we will predict it
46     .G15_0(G47_32), .P15_0(P47_32)
47 );
48
49 //48-63
50 Carry_LookAhead_16bit CLA_16bit_48_63
51 (
52     .Cin(C[48]),
53     .g(g[63:48]), .p(p[63:48]),
54     .C(C[63:48]),.Cout(), //won't use C64 / Cout we will get it from next level
55     .G15_0(G63_48), .P15_0(P63_48)
56 );
57
58 //Layer2=====
59 Carry_LookAhead_4bit CLA_4bit_for_Carry_Prediction
60 (
61     .Cin(Cin),
62     .g0(G15_0), .g1(G31_16), .g2(G47_32), .g3(G63_48),
63     .p0(P15_0), .p1(P31_16), .p2(P47_32), .p3(P63_48),
64     .C1(C[16]), .C2(C[32]), .C3(C[48]), .Cout(Cout), //Get C64 / Cout from this level
65     .G3_0(), .P3_0() //no need for Group Signal, last level reached
66 );
67
68 //=====
69
70 //3.SUM Evaluation=====
71 assign S = C ^ p; //this easy :)
72
73
74 endmodule
```

Testbench *[100 Random Tests]*

```
1 module CLA_64bit_tb;
2
3     reg [63:0] A,B; //Inputs are regs
4     reg cin;
5
6     wire [63:0] sum; //Outputs are wires
7     wire cout;
8
9     //Instantiate
10    Carry_LookAhead_64bit CLAtester (A, B, cin,sum, cout);
11
12    //Task: validation of output
13    task validate_output;
14    |    if({cout,sum} != A + B + cin) $display("Error: Adding A + B + cin, time: %0t", $time);
15    endtask
16
17
18    initial
19    begin
20        A=0; B=0; //initial state
21        cin =0;
22        #10
23
24        repeat(100) //100 randomized tests
25        begin
26            A = {$random, $random}; //$random gives 32 bit only
27            B = {$random, $random};
28            cin = $random % 2; // ensure 0/1
29
30            #5; //wait for output
31
32            validate_output();
33        end
34
35        $display("Test case 1:"); //Test 1
36        A = 64'd7; B = 64'd4; cin = 0;
37
38        #10 //wait for out and monitor
39        if(sum == 64'd11 && (!cout) ) $display("Test 1 pass, time: %0t",$time);
40        else $display("Test 1 fail");
41
42
43        #10
44        $stop;
45    end
46
47    initial //monitor i/o
48    begin // this event is in the postponed event region
49        $monitor("Time=%0t | A=%b | B=%b | cin =%b | Sum=%b | Cout=%b",
50        $time,A,B,cin,sum,cout);
51    end
52
53 endmodule
```

-Waveforms , Results

A	64'd16372...	0	1303060099158949505	5106969336329962597	4261515162043665725	16372595406958724293	98861995006016120463
B	64'd15353...	0	9548992447060661859	9887461744194085633	8562565468134438796	15353847184146560997	8732596169457702606
cin	1'h1						
sum	64'd13279...	0	10851998546219611365	14994431080524048231	12824080630178104522	13279698517395733675	172051601784271453
cout	1'h1						

A	64d16341...	16341960806660327772	12821574841579833098	14901510245151948438	3864050460875172565	16700881026851031331	8748451124407591306
B	64d16036...	16036800117854656557	13851703012334248224	9908965999149594637	9301984931420192430	362932342453635786	2361209200367580376
cin	1h1						
sum	64d13932...	13932016850805432714	8226533780204529706	6363732170591991460	13166035392295364996	17063813369304667117	11109660324775171682
cout	1h1						

	64d16341...	4254660808000253417	13868345221686599618	12899112415840156079	17835792670034704353	17481952193671129640	13527778052692841838	7
	64d16036...	5035740762486234881	16331684889271597616	17642973489315033080	14606436475012792046	3616246894599354153	4659806700656638421	4
	1h1							
	64d13932...	9290401570486488299	11773286037248645619	12095341831445637543	13995485071337944783	2651455014560932178	18187584753349480259	11
	1h1							

[illegible]

-No Errors

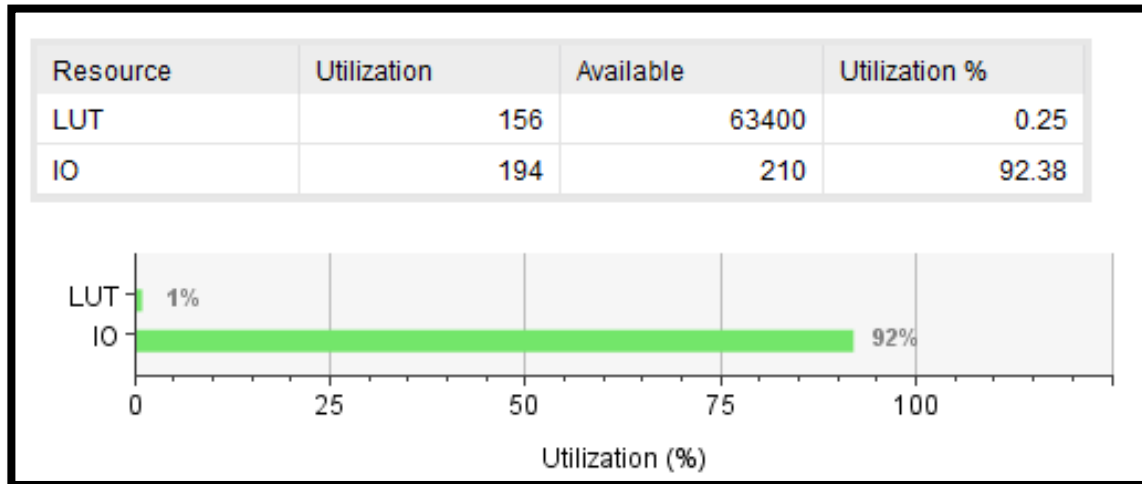
```
# Test case 1:  
# Time=510 | A=00000000000000000000  
# Test 1 pass, time: 520
```

-Pass

FPGA Implementation

[xc7a100tcsg324-1]

-Utilization Report



-Timing Report

Worst Negative Slack (WNS):	-9.747 ns
Total Negative Slack (TNS):	-564.030 ns
Number of Failing Endpoints:	65
Total Number of Endpoints:	65

Given: 5ns max delay ∴ Critical Delay = 14.747ns

Report on limitations

-This Architecture using 4 bit CLAs is limited to (4^n) bits addition